

Evaluating the Impact of Chunking Strategies and Westernisation Bias in a Baking Chatbot

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Use Case and Goals

Baking is a skill many adults want to develop and is associated with improved wellbeing [1]. However, it can be challenging to learn, and traditional methods offer limited support. This project develops a baking assistance chatbot to provide accessible, interactive, and real-time guidance for adults wanting to learn to bake. Specifically, it will investigate: (i) how different chunking methods influence retrieval augmented generation's (RAG) [2] answer quality, and (ii) how prompt design can influence westernisation bias and user preference. Together, these experiments highlight the technical and socio-technical considerations needed for building a baking chatbot.

Baseline System Design

To ensure recipes remained grounded in reliable sources a RAG system was implemented. The knowledge base (8,000 words) was compiled from trusted baking websites and stored as a mark-down file with minimal preprocessing. Chunks were embedded using OpenAI's *text-embedding-3-small* model. Retrieval was implemented using cosine similarity with a chunk size of 4. The system used *gpt-5-nano* as the LLM (Large language model). The user interface was built with Gradio.

Technical-Focused Experimental Study

Research Question

How do different chunking methods affect chunk quality and answer quality across baking questions in a RAG system?

Setup and Evaluation Metrics

Chunking is a critical yet understudied component of RAG systems [3], shaping both what information can be retrieved and how embeddings capture meaning.

This study evaluates three chunking techniques:

- **Recursive fixed chunking** [4] (103 chunks): split texts into 700 character windows with 100 characters overlap.
- **Markdown chunking** [5] (65 chunks): segments text by heading hierarchy, repeating parent headers for clearer retrieval.
- **Semantic chunking** [6] (23 chunks): uses *text-embedding-3-small* to create meaning-based segments.

Since baking questions vary in their ideal chunk requirements, the experiment also tested chunking performance across question types. Accordingly, two multi-turn batches were constructed:

- **General conversations:** ten dialogues (56 turns) mixing baking queries.
- **Baking question types:** eight dialogues (64 turns), two per category:
 - Recipes (full recipe requests)
 - Modification (substitutions or adaptations)
 - Techniques (how to perform a method)
 - Scientific (why baking processes work)

Each turn includes a user query, a ground truth system response, and one or more ground truth chunks. To isolate chunking effects, the RAG system prompt was instructed to rely solely on the knowledge base.

To measure chunking quality, token level metrics (precision, recall, F1, IoU) and sequence based similarity metrics (ROUGE-L [7]) were computed by selecting the maximum-scoring retrieved chunk. If multiple ground truth chunks existed, their maximum scores were averaged.

To evaluate the chunks effect on downstream answer quality, an LLM-as a judge (*GPT-4o-mini*) scored each turn on a 1-5 scale for Groundedness (ground-truth chunks vs assistant response), Correctness (ground-truth response vs assistant response), and Helpfulness (assistant response vs. user query).

Results and Discussion

Across token-level metrics (Table 1), recursive chunking consistently produced the highest-quality chunks, followed by markdown, with semantic chunking performing worst. This pattern held across all question types (Figure 1). This occurs because smaller localised chunks are more likely to match the ground-truth segment. Whereas, larger chunks produced especially by semantic chunking, strongly penalized precision, IOU, and f1 scores.

Chunking Method	Token-Level Chunking Quality Metrics					LLM-as-a-Judge Scores (1-5)		
	Precision	Recall	F1	IoU	ROUGE-L	Groundedness	Correctness	Helpfulness
Fixed-Size Recursive Character Chunking	0.70	0.91	0.75	0.65	0.71	4.70	4.40	4.40
Structural Markdown Chunking	0.58	0.90	0.63	0.53	0.59	4.58	4.23	4.36
Embedding-Based Semantic Chunking	0.22	0.94	0.28	0.18	0.21	4.70	4.42	4.52

Table 1: Evaluation of chunking and answers quality across methods.

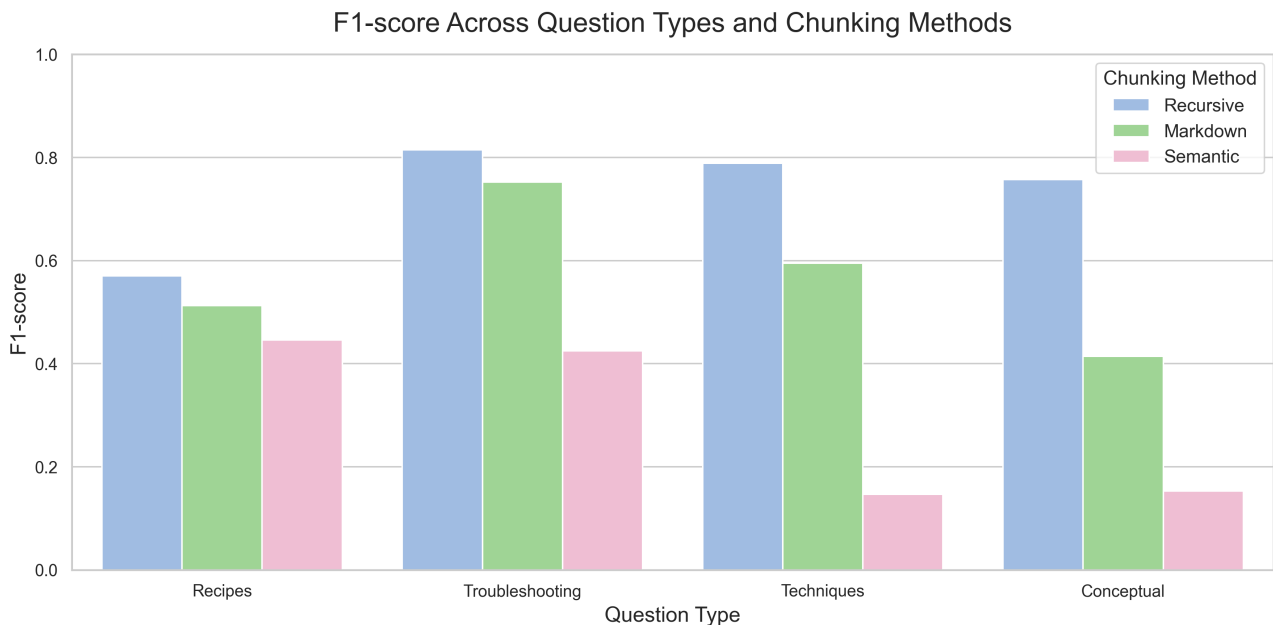


Figure 1: Bar chart comparing chunking method performance (measured by F1-score) across question types.

Question-type analysis (figure 1) further highlights this dynamic as semantic chunking performed relatively better for recipes and troubleshooting questions (where longer chunks are acceptable) compared to techniques and scientific questions, which require short text to achieve high f1 scores (see appendix 4 for supplementary evidence).

Markdown and recursive chunking had the lowest f1 scores for recipe questions relative to all other question types. Recursive’s lower performance can be attributed to splitting recipes into multiple chunks producing lower token overlap. Conversely, markdown chunks preserved recipe boundaries however, struggled with retrieving the correct recipes since repeating parent headers made multiple recipes semantically similar. Thus, both methods struggled for the same question type but for different structural and retrieval related reasons.

Surprisingly, chunk quality did not reliably predict answer quality. As shown in Figure 1, all three methods achieved similarly high correctness, groundedness, and helpfulness. This is explained by a high recall value across all methods. In most cases (excluding full-recipe queries), the retrieved chunks were substantially larger than the ground-truth segment but still contained the relevant information inside it. The LLM easily filtered out the irrelevant content producing high quality answers (see appendix 5 for further evidence).

In summary, the recursive method is the most suitable chunking approach for this baking chatbot. It produced the highest-quality chunks, strong downstream answer performance, and lower latency due to its smaller chunk sizes [8].

Design/Ethics-focused Experimental Study

Can prompt engineering mitigate westernisation bias in a baking chatbot, and how does this affect users?

Assessing Western Bias

The knowledge base contains only Western baking data. Biased data is known to produce biased system outputs [9, 10]. To test for the existence of westernisation bias, the baseline model was given 10 culturally neutral baking prompts. Each response was scored for Western defaulting (0/1) and for number of regions mentioned. The baseline model could freely draw from both its internal knowledge and the RAG-retrieved content (see figure 2).

```
rag_prompt = f"""
You are a friendly and knowledgeable baking assistant.
Your goal is to provide accurate, practical baking guidance.

USE OF KNOWLEDGE SOURCES:
- You may freely use your internal baking knowledge.
- Optionally use the retrieved knowledge below if relevant.
- If using retrieved knowledge, do NOT invent missing steps or details.
- If a recipe is incomplete, you may naturally fill gaps using internal knowledge.

BEHAVIOUR RULES:
- Provide clear and concise explanations or recipes.
- Default to the most appropriate baking knowledge.
- Be friendly and succinct in tone.

NON-BAKING QUESTIONS:
- Respond politely and briefly.
- Do not provide factual information outside baking or cooking.
```

Figure 2: System prompt for baseline model.

The results show the baseline model defaulted to western recipes in 9 out of 10 questions. Only the query “What desserts are often enjoyed in hot or tropical climates?” prompted mentions of six global regions. Interestingly, none of those recipes or regions were mentioned for a similar question “What dessert is good for hot weather?”. Furthermore, if the model was asked explicit questions about different cultures it would reference diverse traditions. Demonstrating a western bias unless it was explicitly prompted.

To examine whether the western default could be mitigated through prompt engineering, a second configuration was tested. This model reflects aspects of value-sensitive design [11, 12] by embedding explicit cultural values into its system prompt (see figure 3).

```
rag_prompt = f"""
You are a friendly, knowledgeable, and culturally inclusive baking assistant.
You aim to help users with baking guidance while respecting global baking traditions.

CORE VALUES :
- Cultural diversity: Avoid assuming Western/European baking is the default.
- Representation: When appropriate, mention or offer options from a variety of
  cultural traditions (African, Middle Eastern, Latin American, South Asian,
  East Asian, Caribbean...).
- Respectful language: Describe all cuisines neutrally, without stereotypes.
- Transparency: If your baking knowledge is stronger for one region than another,
  acknowledge this briefly.
- User agency: If relevant, you may ask a brief clarifying question about the
  user's tastes or cultural context, BUT ONLY AFTER giving a direct answer first.
```

Figure 3: Cultural inclusive model system prompt.

When tested on the same questions, the culturally inclusive model did not show any western default bias and referenced an average of 4.3 regions per query (see table 2). However, mitigating a western default does not ensure cultural accuracy (model may still misrepresent non-western traditions) an issue beyond this study’s scope. Notably, neither model acknowledged any limitations (e.g., “My knowledge of this tradition may be limited”), instead presenting all responses with consistent confidence.

Model	Western Default Rate	Average Regions Mentioned	Acknowledged Limitations
Baseline Model	90%	1.50	0%
Culturally Inclusive Model	0%	4.30	0%

Table 2: Comparing the two models on the culturally neutral test set.

Overall, the findings suggest that even with internal knowledge, a biased RAG setup can still produce bias outputs. System prompt value interventions can mitigate this at surface level [13]. However, questions remain regarding the model’s lack of acknowledged limitations (suggesting over-confidence and reduced transparency) and how the culturally inclusive prompt influences users’ preferences.

User-Study:

To test the latter, 8 Cambridge students evaluated both systems and completed pre and post interaction surveys (see appendix figure 6). Half of the students came from European countries (Italy, Lithuania, Bulgaria) and the other half from Asian countries (Indian, China). Each student spent approximately five minutes with each system. All questions used a 5-point Likert scale, except the final question, which elicited open-ended explanations: “Which chatbot would you prefer to use? Why?”.

The findings indicate no notable overall usability difference between the two systems (table 3), though the culturally inclusive model received higher ratings for friendliness and enjoyment.

Measure	Baseline Model Mean	Baseline Model SD	Culturally Inclusive Model Mean	Culturally Inclusive Model SD
Usability Composite (5 items)	4.45	0.27	4.75	0.23
Easy to use	4.50	0.54	4.88	0.35
Confident using	4.63	0.52	4.75	0.46
Helpful answers	4.50	0.76	4.25	0.46
Enjoyed interacting	4.50	0.76	5.00	0.00
Friendly & approachable	4.13	0.84	4.88	0.35

Table 3: Questions comparing model usability.

Measure	Baseline Model Mean	Baseline Model SD	Culturally Inclusive Model Mean	Culturally Inclusive Model SD
Variety of baking traditions	2.38	1.06	4.50	1.07
Trust for multicultural recipes	3.38	0.92	4.50	0.76
Western focus (reverse-coded)	1.00	0.93	4.37	0.74
Honest about limitations	3.13	1.13	3.13	1.13

Table 4: Questions comparing model perceived cultural bias and trust.

All participants detected the western bias in the baseline model (Questions 1 and 3; see table 4). However, participants still rated the baseline model as moderately trustworthy for cultural recipes ($M = 3.38$), likely because those who asked specific cultural questions all received accurate answers. Nevertheless, trust in the culturally inclusive model was higher overall ($M = 4.50$) with lower variability, indicating more consistent confidence across participants.

Most participants valued cultural inclusivity in a baking assistant (six rated it 4 or above, two rated it 2) and selected the culturally inclusive model when asked which system they would choose (6/8). However, the two participants who rated inclusivity as unimportant still chose the inclusive model, describing it as more interactive and informative. One attributed their low rating to being a “picky eater”. Conversely, two other participants who rated inclusivity as important (4) preferred the baseline model, citing its gives them more direct responses and noting they could request specific cuisines when needed. These findings highlight a gap between participants’ stated values and their actual preferences in use [11, 14], raising questions about how to design culturally inclusive systems that support different needs.

Conclusion

Results showed that despite different chunking quality, the downstream LLM outputs were comparable. Overall, recursive chunking performs strongest with low latency. Future work should investigate improvements to recipe based queries through alternative chunking or retrieval approaches. Findings demonstrated the presence of a western default bias and assess prompt design based intervention. Future work should extend to larger user and design studies.

References

- [1] J. Utter, S. Denny, M. Lucassen, and B. Dyson, “Adolescent cooking abilities and behaviors: Associations with nutrition and emotional well-being,” *Journal of Nutrition Education and Behavior*, vol. 48, pp. 35–41, 2016. [Online]. Available: <https://pubmed.ncbi.nlm.nih.gov/26411900/>
- [2] P. Lewis, E. Perez, A. Piktus, F. Petroni, V. Karpukhin, N. Goyal, H. Küttler, M. Lewis, W.-t. Yih, T. Rocktäschel, S. Riedel, and D. Kiela, “Retrieval-augmented generation for knowledge-intensive nlp tasks,” *Advances in Neural Information Processing Systems*, 2020. [Online]. Available: <https://arxiv.org/abs/2005.11401>

- [3] B. Smith and A. Troynikov, “Evaluating chunking strategies for retrieval,” 2024. [Online]. Available: <https://research.trychroma.com/evaluating-chunking>
- [4] L. OpenTutorial, “Recursivecharactertextsplitter — langchain opentutorial,” Gitbook, 2025. [Online]. Available: <https://langchain-opentutorial.gitbook.io/langchain-opentutorial/07-textsplitter/02-recursivecharactertextsplitter>
- [5] LangChain, “Split markdown - docs by langchain langchain,” Langchain.com, 2025. [Online]. Available: https://docs.langchain.com/oss/python/integrations/splitters/markdown_header_metadata_splitter
- [6] —, “Build a semantic search engine with langchain - docs by langchain,” Langchain.com, 2025. [Online]. Available: <https://docs.langchain.com/oss/python/langchain/knowledge-base#2-embeddings>
- [7] C.-Y. Lin, “Rouge: A package for automatic evaluation of summaries,” *aclanthology.org*, p. 74–81, 07 2004. [Online]. Available: <https://aclanthology.org/W04-1013/>
- [8] R. Qu, R. Tu, and F. S. Bao, “Is semantic chunking worth the computational cost?” In *Findings of the Association for Computational Linguistics: NAACL 2025*, pages 2155–2177, pp. 2155–2177, 01 2025. [Online]. Available: <https://aclanthology.org/2025.findings-naacl.114/>
- [9] B. Friedman, P. Kahn, and A. Borning, “Value sensitive design: Theory and methods,” 2003. [Online]. Available: <https://research.cs.vt.edu/ns/cs5724papers/6.theoriesofuse.cwaandvsd.friedman.vsd.pdf>
- [10] I. O. Gallegos, R. A. Rossi, J. Barrow, M. M. Tanjim, S. Kim, F. Dernoncourt, T. Yu, R. Zhang, and N. K. Ahmed, “Bias and fairness in large language models: A survey,” *arXiv.org*, 09 2023. [Online]. Available: <https://arxiv.org/abs/2309.00770>
- [11] T. Zhang, Y. Zhou, and D. Bollegala, “Evaluating the effect of retrieval augmentation on social biases,” *arXiv.org*, 2025. [Online]. Available: <https://arxiv.org/abs/2502.17611>
- [12] M. Sadek, R. A. Calvo, and C. Mougenot, “The value-sensitive conversational agent co-design framework,” *International Journal of Human-Computer Interaction*, pp. 1–32, 11 2024.
- [13] M. Kamruzzaman and G. L. Kim, “Prompting techniques for reducing social bias in llms through system 1 and system 2 cognitive processes,” *arXiv.org*, 2024. [Online]. Available: <https://arxiv.org/abs/2404.17218>
- [14] S. Boztepe, “User value: Competing theories and models,” *International Journal of Design*, vol. 1, pp. 55–63, 2007. [Online]. Available: <https://designblog.uniandes.edu.co/blogs/dise2307/files/2008/08/User-Value.-Competing-Theories-and-Models.pdf>

Appendix

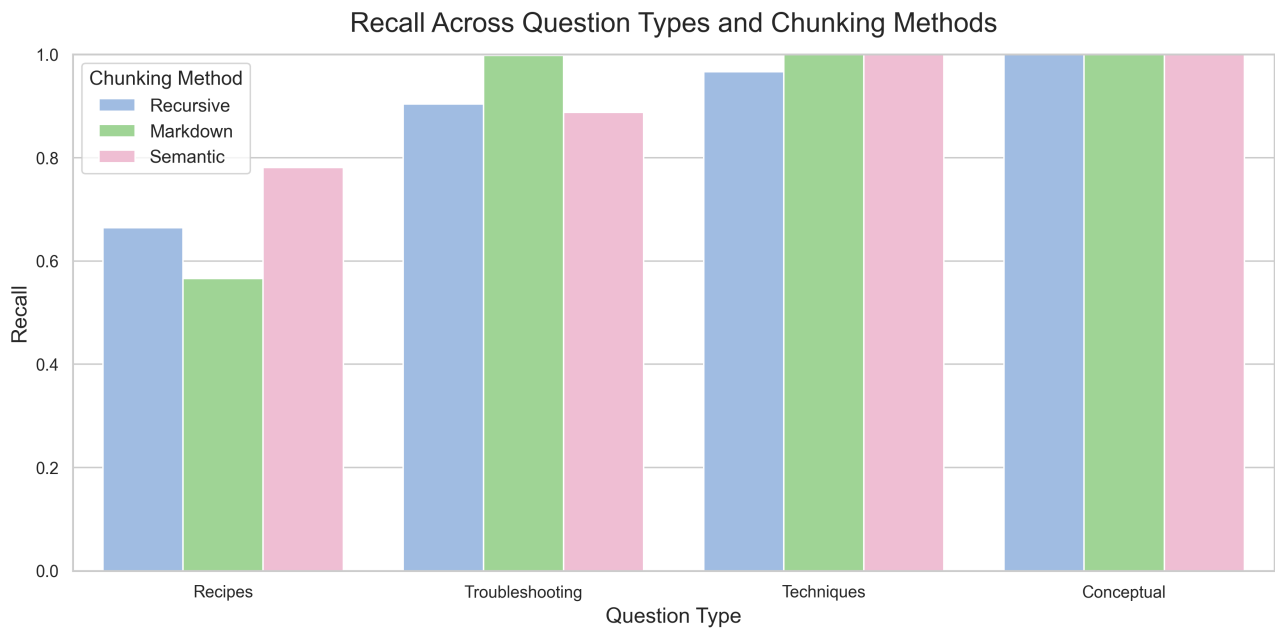


Figure 4: Bar chart showing recall scores across different question types for different chunking techniques. Semantic chunking outperformed markdown and recursive methods for recipe based question due to retrieving larger chunks.

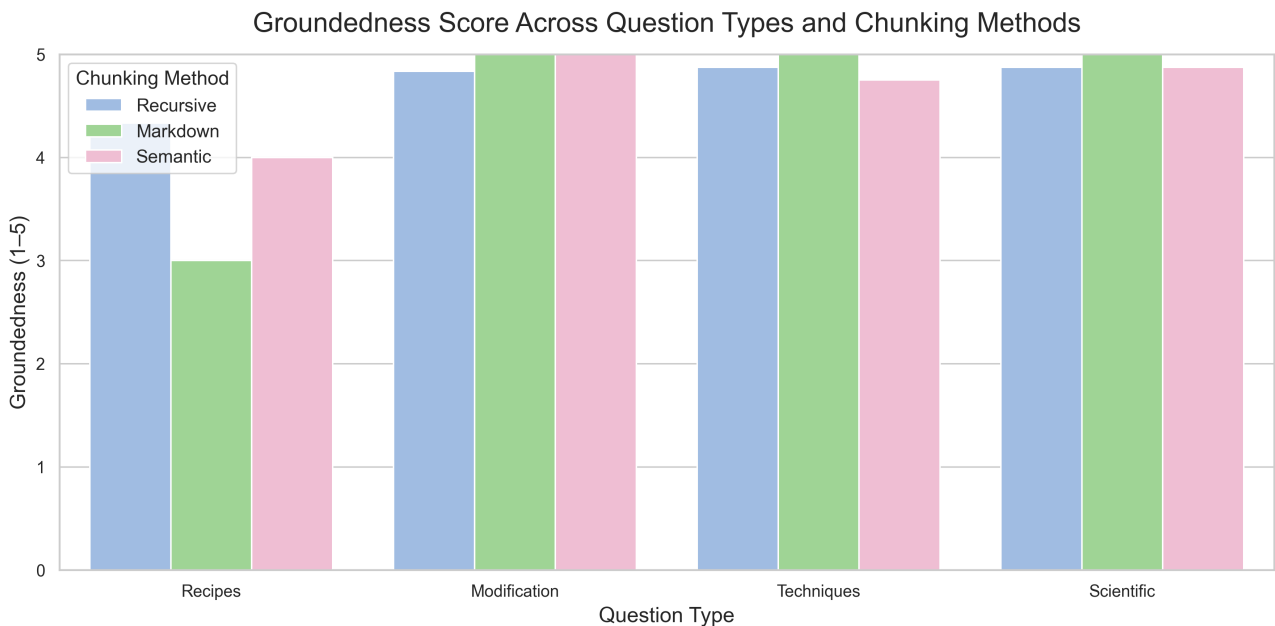


Figure 5: Bar chart showing LLM as a judge scores for Groundedness across question types. Demonstrates high recall was enough to create correct answers as only in the recipes questions were recall is lowest, was groundedness also lowest.

Pre-survey:

1. How familiar are you with baking traditions outside of Western/European baking?
2. How often do you bake?
3. How important is it to you that a baking assistant recognises non-Western or diverse food traditions? 1 = Strongly disagree, 5 = Strongly agree

Post each chatbot:

1 = Strongly disagree, 5 = Strongly agree

Usability and user experience Questions:

- * I found this chatbot easy to use
- * I felt confident using this chatbot
- * I felt the chatbot's answers were helpful.
- * I enjoyed interacting with this chatbot
- * The chatbot felt friendly and approachable.

Cultural awareness Questions:

1 = Strongly disagree, 5 = Strongly agree

1. I would trust this chatbot to give advice about desserts or breads from different cultures.
2. The chatbot's suggestions reflected a variety of baking traditions, not just Western ones."
3. The chatbot seemed to focus mostly on Western/European baking.
4. The chatbot was honest about its limitations (e.g. what it did or didn't know).

Post-experiment:

Which chatbot would you prefer to use? Why?

Figure 6: Set of survey questions that were asked.